Four diagonals for doubled letters, five for tripled: completeness of Parikh matrices

Christopher Yamba

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Abstract

The Parikh matrix of a word over an ordered alphabet records, diagonal by diagonal, the numbers of occurrences of the scattered subwords made of consecutive letters. For permutations the pair counts on the third diagonal determine the whole matrix (the completeness theorem of the author's 2006 thesis, and a special case of Salomaa's γ-property theorem of 2005), while Salomaa showed that in general no fixed number of diagonals suffices. We prove that if every letter other than the first and the last occurs at most twice, the first four diagonals determine the Parikh matrix, and that if they occur at most three times, the first five do; both results come with explicit formulas. The second follows from a local pivot theorem: the number of occurrences of $j(j+1)\cdots l$ is determined by the shorter such counts as soon as one letter at distance at least 2 from j and from l occurs at most twice, or it and its two neighbours occur at most three times. The proof factorises a block of the Parikh matrix through the copies of that letter and uses a finite chain lemma. Explicit words show that the distance condition and the neighbour bounds cannot be relaxed, and that the pivot may not occur five times; whether it may occur four times is open. In Salomaa's sharpness construction one copy of the first letter suffices: for every $n \geq 2$ two words of content $(1, n, \ldots, n)$ have Parikh matrices that differ only in the top right entry, with ballot numbers, among them the Catalan number C_n, as other entries. We conjecture that counts of length at most $n+1$ suffice when the inner letters occur at most n times; this holds for $n \leq 3$, and exhaustive computation supports it for $n = 4$. Both completeness theorems are formalised in Lean 4.

Keywords: Parikh matrix; scattered subword; M-ambiguity; Cauchy inequality for words; total positivity; ballot numbers; formal verification.
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1 Introduction

Let w be a word over the ordered alphabet $1 < 2 < \cdots < k$. Its *Parikh matrix* $\Psi(w)$, introduced by Mateescu, Salomaa, Salomaa and Yu [4], is the upper unitriangular $(k+1) \times (k+1)$ matrix whose entry $(j, l+1)$ is the number $|w|_{j..l}$ of occurrences of $j(j+1)\cdots l$ in w as a scattered subword. Its second diagonal is the classical Parikh vector, the third holds the pair counts $|w|_{i(i+1)}$, the fourth the triple counts $|w|_{i(i+1)(i+2)}$, and so on: the d-th diagonal holds the counts of length $d-1$.

The author's 2006 thesis [14] read the Parikh matrix of a permutation diagonal by diagonal ("it seems natural to continue this procedure, investigating the entries by going through the diagonals in increasing order" [14, printed p. 8, PDF p. 12]) and proved the *Parikh matrix completeness theorem*: the third diagonal of the Parikh matrix of a permutation determines the whole matrix [14, Theorem 3.5, printed pp. 19–20, PDF pp. 23–24]. For words with repeated letters the third diagonal no longer suffices [9, p. 226], and [15, Theorem A] says exactly when it does. Salomaa's γ-property [8], in the form quoted in [9, Theorem 3, p. 226], gives a sufficient condition for the second and third diagonals to determine the matrix; it

appeared the year before the thesis and already contains the permutation case, since the condition is empty when every letter occurs once; its condition fails as soon as two adjacent letters occur at least twice each and have a pair count other than 0, 1, $n_i n_{i+1} - 1$ and $n_i n_{i+1}$. Salomaa then showed that *no* fixed number of diagonals suffices in general: for every entry there are two Parikh matrices that differ only there [9, Theorem 4, p. 228]. He adds that it is "not so easy to give examples of different Parikh matrices whose second, third and fourth diagonals coincide. (Indeed, it was sometimes conjectured that such examples do not exist.)" [9, p. 228]. His examples use many repeated letters: in his witnesses for a window of $n+1$ letters, the inner and last letters occur n times and the first letter 2^{n-1} times (and $2^{n-1}t$ times when the two corners are to differ by t) [9, Lemma 3, p. 227, and proof of Theorem 4].

This note asks what happens when the multiplicities are bounded, and answers the first two cases. For $j < l$ the *window* $[j, l]$ is the set of letters $j, \dots, l$, its *corner* is the count $|w|_{j..l}$, and $j+1, \dots, l-1$ are its *inner letters*.

Theorem A. *Let $j < l$ with $l - j \geq 3$, and let w be a word in which every inner letter of $[j, l]$ occurs at most twice. Then $|w|_{j..l}$ is determined by the counts $|w|_{j'..l'}$ with $j \leq j' \leq l' \leq l$ and $l' - j' < l - j$: every word w' with the same such counts has $|w'|_{j..l} = |w|_{j..l}$. The value is given explicitly by Theorem 4.2.*

By induction on the length of the window we obtain the completeness theorem for doubled letters.

Corollary 1.1. *If every letter other than 1 and k occurs at most twice in w, then the first four diagonals of $\Psi(w)$ determine $\Psi(w)$: every word with the same first four diagonals has the same Parikh matrix.*

For permutations it is weaker by one diagonal than the thesis's theorem, and it contains Simpson's result for palindromes in which every letter occurs exactly twice [13, Corollary 4, p. 331], on which the third diagonal is a function of the second [13, Theorem 3(b)]. It confirms, for these words, the statement reported in [9, p. 228]. Neither parameter of Corollary 1.1 can be improved: the first three diagonals do not suffice (Example 4.5), and with four diagonals "twice" cannot be replaced by "three times", by Salomaa's construction [9, Lemma 3 and proof of Theorem 4] with $n = 3$, or by Theorem B (Example 4.5). The proof (Sections 3 and 4) rests on two rules that extend the cut rule (Lemma 2.2), the product form of the rule behind the thesis's completeness theorem. The *rectangle rule* (Lemma 3.1) factorises a long count across any edge whose two letters meet as a rectangle; it gives a sufficient condition for equality in the Cauchy inequality for words [5, 7]. The *crossing chain* (Proposition 4.1) computes a long count along interleaved pairs $i(i+1)i(i+1)$. In particular the Parikh classes of the words in which each of k letters occurs twice are the sets of words with given third and fourth diagonals (Corollary 4.6).

In the other direction, the following family shows that the first letter of Salomaa's witnesses can occur just once.

Theorem B. *For every $n \geq 2$ there are two words A_n, B_n of content $(1, n, n, \dots, n)$, over the letters $1, \dots, n+1$, whose Parikh matrices differ only in the corner entry: $|A_n|_{1..(n+1)} = 1$ and $|B_n|_{1..(n+1)} = 0$. For $2 \leq i < j \leq n+1$ and $m = j - i$, the counts $|A_n|_{i..j} = |B_n|_{i..j}$ are the ballot numbers $\binom{n+m}{m+1} - \binom{n+m}{m-1}$; in particular $|A_n|_{2..(n+1)}$ is the Catalan number C_n.*

So, as Salomaa's construction already shows, when the inner letters occur n times the counts of length at most n do not determine the corner; Theorem B shows that this needs only one copy of the first letter. We conjecture that this is the only obstruction.

Conjecture C. Let $n \geq 1$. If every inner letter of $[j, l]$ occurs at most n times in w and $l - j \geq n + 1$, then $|w|_{j..l}$ is determined by the shorter counts of the window. Hence, if every letter other than 1 and k occurs at most n times, the first $n + 2$ diagonals determine the Parikh matrix.

The case $n = 1$ is the cut rule of Lemma 2.2, and $n = 2$ is Theorem A. Our second main result proves the case $n = 3$. It pivots on a letter instead of an edge, and it is local: it needs a bound on one letter and its two neighbours only.

Theorem D. *Let $j < l$ with $l - j \geq 4$, and let m be a letter with $j + 2 \leq m \leq l - 2$. Suppose that m occurs at most twice in w, or that $m - 1$, m and $m + 1$ each occur at most three times. Then $|w|_{j..l}$ is determined by the shorter counts of the window $[j, l]$. In the first case the multiplicities of the other letters are arbitrary. The value is given explicitly by Theorem 6.8.*

Corollary 1.2. *Conjecture C holds for $n = 3$. In particular, if every letter other than 1 and k occurs at most three times in w, then the first five diagonals of $\Psi(w)$ determine $\Psi(w)$.*

Corollary 1.3. *If every letter m with $3 \leq m \leq k - 2$ occurs at most twice in w, then the first five diagonals of $\Psi(w)$ determine $\Psi(w)$, whatever the multiplicities of the letters 1, 2, $k - 1$ and k.*

Unlike Corollary 1.1, Corollary 1.2 is not a case of the statement reported in [9, p. 228]: when letters occur three times, four diagonals do not suffice (Example 4.5), and with five diagonals "three times" cannot be replaced by "four times" (Theorem B with $n = 4$).

Every occurrence of $j(j + 1) \cdots l$ passes through exactly one letter m, so the block $(|w|_{i..l'})_{i \leq m \leq l'}$ of $\Psi(w)$ has rank at most the multiplicity of m (Lemma 6.1). A 2×2 minor decides whether the letters m meet their neighbours as a rectangle (Lemma 6.2, known in substance); a 3×3 minor then gives the corner unless three letters m lie in a *rigid* position (Lemmas 6.4 and 6.6); and a finite chain lemma (Lemma 6.6) identifies the rigid positions from three short counts when the three letters occur at most three times. These minors count families of vertex-disjoint paths in a planar network, as in total positivity [1] (Remark 6.5). The hypotheses of Theorem D cannot be relaxed, except possibly at a pivot that occurs four times (Proposition 7.1): for instance, there are two words of content $(1, 3, 3, 4, 8)$ whose Parikh matrices differ only in the corner, although, at the pivot $m = 3$, the letters 2 and 3 occur three times and the letter 4 occurs four times.

Section 7 reports exhaustive evidence for $n = 4$; if true, the conjecture says that Salomaa's construction is optimal in its inner multiplicities. Section 8 describes a formalisation in Lean 4 of Theorems A and D, Corollaries 1.1 to 1.3, the rectangle rule, the crossing chain, the chain lemma and the corner statement of Theorem B for $n = 3, 4$, together with the computer checks of every numerical statement; the supplementary files are on the article's web page and in the repository `https://github.com/cyamba/yamba-ai-math/tree/main/research/parikh`. Our companion articles [15, 16] study the largest Parikh classes and the values of the fourth diagonal for words with repeated letters, and [15] determines exactly when the first three diagonals suffice [15, Theorem A]; the present note takes up the question of the further diagonals raised at the end of [15]. Its preliminaries, the product formula (2.1), the realisation of projections and the cut rule (Lemma 2.2) are repeated from [15, Section 2 and Lemma 3.1] to keep the note self-contained; no proof here depends on [16, 15].

What is new in version 2

Version 1 of this note (23 October 2026), titled *Four diagonals suffice: completeness of Parikh matrices when letters occur at most twice*, contained Theorems A and B, Corollaries 1.1 and 4.6 and Conjecture C, with its sections numbered 1–7. Version 2 (24 October 2026) changes the following.

- *New results:* Theorem D (the pivot theorem) and its explicit form Theorem 6.8; Corollary 1.2, which proves Conjecture C for $n = 3$; Corollary 1.3; and Proposition 7.1, explicit words showing that the hypotheses of Theorem D cannot be relaxed, except possibly at a pivot that occurs four times.
- *New section:* Section 6, with the proofs and the chain lemma (Lemma 6.6, Table 1).
- *Renumbering:* Sections 6 and 7 of version 1 are now Sections 7 and 8, and its Table 1 is now Table 2; all earlier numbers are unchanged.
- *Corrections and additions:* Remark 3.2 now says that the equality case is known in full for single letters x, y, z (version 1 said "for y a single letter"); Example 4.5 prints Salomaa's words; the evidence for $n = 3$ in Table 2 is now a theorem; the Lean formalisation and the supplementary checks cover the new results; and Remark 6.5 adds the planar-network picture of total positivity [1, 2].

The title has changed accordingly. Corrections, comments, questions and feedback are welcome at `christopher.yamba@yambacode.com`.

2 Preliminaries

A *content* is a vector $\mathbf{n} = (n_1, \dots, n_k)$ of positive integers, and $\mathcal{W}(\mathbf{n})$ is the set of words over $1 < \cdots < k$ with n_i letters i. Subwords are scattered: $|w|_u$ counts the position sequences $p_1 < \cdots < p_\ell$ with $w_{p_1} \cdots w_{p_\ell} = u$. We write $|w|_{j..l} = |w|_{j(j+1)\cdots l}$ for $j \le l$, so that $|w|_{m..m} = n_m$, and call $l - j + 1$ the *length* of the count. The Parikh matrix $\Psi(w)$ has entry $|w|_{j..l}$ in position $(j, l+1)$, and is the product, from left to right, of the matrices $I + E_{m,m+1}$ over the letters m of w [4, Definition 1.1, Theorem 2.1]; in the thesis it is $\psi_{M_k}(w)$ [14, Definition 2.8, Theorem 2.10, printed pp. 12–13, PDF pp. 16–17]. *Edge* i joins the letters i and $i+1$, and $x_i = |w|_{i(i+1)}$ is its *area*. In a window $[j, l]$ the *inner edges* are the edges e with $j < e$ and $e + 1 < l$. A *Parikh class* is the set of words with a given Parikh matrix.

The *projection* of w on edge i, its subword on the letters i and $i+1$, is encoded by $A^{(i)}(w) = (A_1^{(i)}, \dots, A_{n_{i+1}}^{(i)})$, where $A_s^{(i)}$ is the number of letters i before the s-th letter $i+1$; it is nondecreasing, with values in $\{0, \dots, n_i\}$. Let $\mathbf{B}_i$ be the $n_i \times n_{i+1}$ matrix with entry 1 in position (r, s) if the r-th letter i precedes the s-th letter $i+1$, that is, if $r \le A_s^{(i)}$, and 0 otherwise. It is a *Ferrers matrix*: its s-th column is the indicator of the rows $1, \dots, A_s^{(i)}$, the *heights*. Let $\mathbf{1}$ be the all-ones column vector. Expanding the product below, the term indexed by $(r_j, \dots, r_l)$ is 1 exactly when the r_j-th letter j, …, the r_l-th letter l form an occurrence of $j(j+1)\cdots l$; hence

$$|w|_{j..l} = \mathbf{1}^\top \mathbf{B}_j \mathbf{B}_{j+1} \cdots \mathbf{B}_{l-1} \mathbf{1} \qquad (j < l). \tag{2.1}$$

Conversely, every tuple of nondecreasing sequences $A^{(i)} \in \{0, \dots, n_i\}^{n_{i+1}}$ is the tuple of projections of a word: start from 1^{n_1} and insert, for $i = 1, \dots, k-1$ in turn, the s-th letter $i+1$ just before the $(A_s^{(i)} + 1)$-th letter i, or at the end if $A_s^{(i)} = n_i$ (so every area $0 \le x_i \le n_i n_{i+1}$ occurs, which is [5, Lemma 8, p. 19]).

Lemma 2.1. *Let $\mathbf{B}_e$ be the Ferrers matrix of edge e in w.*

(i) *The rank of $\mathbf{B}_e$ is the number of distinct nonzero heights.*

(ii) *$\mathbf{B}_e$ has rank at most 1 if and only if w has no subword $e(e+1)e(e+1)$.*

(iii) *If $n_e, n_{e+1} \le 2$, then $\mathbf{B}_e$ has rank 2 if and only if $n_e = n_{e+1} = 2$ and $x_e = 3$, that is, if and only if the letters e, $e+1$ appear in the order $e(e+1)e(e+1)$.*

Proof. (i) Equal heights give equal columns and zero heights zero columns, and the indicators of initial segments of distinct lengths are linearly independent. (ii) An occurrence of $e(e+1)e(e+1)$ uses the r_1-th and r_2-th letters e and the s_1-th and s_2-th letters $e+1$ with $r_1 \le A^{(e)}_{s_1} < r_2 \le A^{(e)}_{s_2}$, so two distinct nonzero heights occur; conversely, from nonzero heights $A^{(e)}_{s_1} < A^{(e)}_{s_2}$ take $r_1 = 1$ and $r_2 = A^{(e)}_{s_2}$. (iii) A Ferrers matrix with at most two rows and two columns has two distinct nonzero heights only if it is 2×2 with heights $(1, 2)$, which is the only 2×2 Ferrers matrix of area 3. □

Lemma 2.2 (Cut rule)**.** (i) *If $j \le e < e+1 \le l$ and $x_e = 0$, then $|w|_{j..l} = 0$.*

(ii) *If $j < m < l$ and $n_m = 1$, then $|w|_{j..l} = |w|_{j..m} \cdot |w|_{m..l}$.*

Proof. (i) $\mathbf{B}_e = 0$ in (2.1). (ii) The row vector $\mathbf{1}^\top \mathbf{B}_j \cdots \mathbf{B}_{m-1}$ and the column vector $\mathbf{B}_m \cdots \mathbf{B}_{l-1}\mathbf{1}$ have one entry each, equal to $|w|_{j..m}$ and $|w|_{m..l}$. □

For permutations, (ii) gives $|w|_{j..l} = \prod_{i=j}^{l-1} |w|_{i(i+1)}$, the product form of the rule behind the thesis's completeness theorem, that a count is 1 exactly when the two shorter overlapping counts are 1 [14, Proposition 3.4, Theorem 3.5, printed pp. 17–20, PDF pp. 21–24]; with $l = j + 2$, (ii) is the case $|w|_b = 1$ of the equality condition in [5, Lemma 2, p. 12]. So Conjecture C holds for $n = 1$: a window with $l - j \ge 2$ and a single copy of each inner letter is determined by (ii).

3 Rectangles

Lemma 3.1 (Rectangle rule)**.** *If w has no subword $e(e+1)e(e+1)$, then for all $j \le e < e+1 \le l$,*

$$|w|_{j..l} \cdot |w|_{e(e+1)} = |w|_{j..e+1} \cdot |w|_{e..l}.$$

Proof. By Lemma 2.1(ii), $\mathbf{B}_e = uv^\top$ for column vectors u, v. With $P = \mathbf{1}^\top \mathbf{B}_j \cdots \mathbf{B}_{e-1} u$ and $Q = v^\top \mathbf{B}_{e+1} \cdots \mathbf{B}_{l-1}\mathbf{1}$, (2.1) gives $|w|_{j..l} = PQ$, $|w|_{j..e+1} = P\,(v^\top \mathbf{1})$, $|w|_{e..l} = (\mathbf{1}^\top u)\,Q$ and $|w|_{e(e+1)} = (\mathbf{1}^\top u)(v^\top \mathbf{1})$. □

In words, w avoids $e(e+1)e(e+1)$ exactly when its letters e and $e+1$ appear in the order $(e+1)^{p_1} e^{q_1} (e+1)^{r_1} e^{s_1}$: the pairs of a letter e before a letter $e+1$ form a rectangle, the first q_1 letters e times the middle r_1 letters $e+1$, and every occurrence of $j \cdots l$ through edge e is an occurrence of $j \cdots e$ ending in the rectangle followed by an occurrence of $(e+1) \cdots l$ starting in it. When $n_e = 1$ or $n_{e+1} = 1$, edge e is always a rectangle, and Lemma 3.1 follows from Lemma 2.2(ii) with $m = e$ or $m = e+1$, and is equivalent to it when $x_e \ne 0$.

Remark 3.2. Lemma 3.1 gives a sufficient condition for equality in the Cauchy inequality $|w|_y |w|_{xyz} \le |w|_{xy} |w|_{yz}$ [5, Corollary 4, p. 12], [7, Theorem 4, p. 193], with $y = e(e+1)$, $x = j \cdots (e-1)$ and $z = (e+2) \cdots l$; the condition is not necessary, as $w =$ `1112323444` with $(j, e, l) = (1, 2, 4)$ shows. For single letters x, y, z the equality case is known in full [5, Lemma 2, p. 12]; for a single letter y and longer words x, z it is implicit in the proof of the Cauchy inequality [5, pp. 15–16] (see Remark 6.3). Salomaa notes that "no general theory exists concerning the cases when the Cauchy inequality actually reduces to an equality" [7, p. 193]. For two letters a, b the analogous statement is known: $|w|_{ab} |w|_{a^m b^n} \le |w|_{a^m b} |w|_{ab^n}$ for $m, n \ge 2$, strict exactly when w contains *abab* and the right side is nonzero [7, Theorem 5, p. 195], the case $m = n = 2$ being [5, Lemma 3, p. 12]. We regard Lemma 3.1 as the counterpart of that statement for consecutive letters, not as new in substance. We have not been able to consult [11], whose title and abstract announce a study of equality cases.

4 Crossings, and the proof of Theorem A

Call edge e a *crossing* if $n_e = n_{e+1} = 2$ and $x_e = 3$; by Lemma 2.1(iii) its Ferrers matrix is $U_2 = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and the letters e, $e+1$ appear in the order $e(e+1)e(e+1)$.

Proposition 4.1 (Crossing chain)**.** *Let $j < l$ with $l - j \geq 3$, let every inner letter of $[j, l]$ occur twice and every inner edge be a crossing, and put $p = l - j - 2$. Let a_s be the number of letters j before the s-th letter $j+1$, and d_s the number of letters l after the s-th letter $l-1$ ($s = 1, 2$). Then*

$$|w|_{j..l} = a_1 d_1 + p\, a_1 d_2 + a_2 d_2,$$
$$|w|_{j(j+1)} = a_1 + a_2, \quad |w|_{j..j+2} = 2a_1 + a_2, \quad |w|_{(l-1)l} = d_1 + d_2, \quad |w|_{l-2..l} = d_1 + 2d_2.$$

Proof. By (2.1), $\mathbf{1}^\top \mathbf{B}_j = (a_1, a_2)$ and $\mathbf{B}_{l-1}\mathbf{1} = (d_1, d_2)^\top$, and the p inner edges contribute $U_2^p = \left(\begin{smallmatrix} 1 & p \\ 0 & 1 \end{smallmatrix}\right)$. So $|w|_{j..l} = (a_1, a_2)\, U_2^p\, (d_1, d_2)^\top$. Edge $j+1$ is inner, so $|w|_{j..j+2} = (a_1, a_2)U_2\mathbf{1} = 2a_1 + a_2$; symmetrically for $|w|_{l-2..l}$. □

Combinatorially, an occurrence of $(j+1)\cdots(l-1)$ chooses the first or the second copy of each inner letter, and a crossing forbids the second copy of a letter to be followed by the first copy of the next; so the occurrence switches from first to second copies at most once. One occurrence uses only first copies, one only second copies, and p switch, at one of the p inner edges; this is the matrix U_2^p.

Theorem 4.2. *Let $j < l$ with $l - j \geq 3$, and suppose every inner letter of $[j, l]$ occurs at most twice in w.*

(i) *If some inner edge e has $x_e = 0$, then $|w|_{j..l} = 0$.*

(ii) *Otherwise, if some inner edge e is not a crossing, then $|w|_{j..l} = |w|_{j..e+1} \cdot |w|_{e..l}/x_e$.*

(iii) *Otherwise every inner edge is a crossing, and with $X = |w|_{j(j+1)}$, $Y = |w|_{j..j+2}$, $X' = |w|_{(l-1)l}$, $Y' = |w|_{l-2..l}$ and $p = l - j - 2$,*

$$|w|_{j..l} = (Y - X)(2X' - Y') + p\,(Y - X)(Y' - X') + (2X - Y)(Y' - X').$$

In each case the right side involves only counts $|w|_{j'..l'}$ with $j \leq j' \leq l' \leq l$ and $l' - j' < l - j$, and which case applies is decided by the multiplicities of the inner letters and the areas of the inner edges. The formula in (ii) holds for every inner edge e that is not a crossing; we fix e to be the least one.

Proof. Inner edges exist since $l - j \geq 3$. (i) is Lemma 2.2(i). In (ii), the letters e and $e+1$ occur at most twice and edge e is not a crossing, so $\mathbf{B}_e$, which has at most two rows, has rank at most 1 by Lemma 2.1(iii) and avoids $e(e+1)e(e+1)$ by Lemma 2.1(ii); Lemma 3.1 applies with $x_e \neq 0$, and the two factors are shorter because $j < e$ and $e + 1 < l$. In (iii) every inner letter is an end of an inner edge, so it occurs twice, and Proposition 4.1 applies; solve its last four equations for a_1, a_2, d_1, d_2. The case distinction uses only n_e, n_{e+1} and x_e for the inner edges e. □

Proof of Theorem A. Let w' have the same shorter counts as w in the window. It has the same multiplicities of the inner letters, hence at most two copies of each, and the same areas, so the same case of Theorem 4.2 applies to w and w', with the same right side (in case (ii) with the same edge e, the least inner edge that is not a crossing, which is determined by these counts). □

Proof of Corollary 1.1. Let w' have the same first four diagonals as w. We show $|w'|_{j..l} = |w|_{j..l}$ for $1 \le j \le l \le k$ by strong induction on $l-j$. For $l-j \le 2$ these are entries of the first four diagonals. For $l-j \ge 3$ the inner letters of $[j,l]$ lie in $\{2,\dots,k-1\}$, so they occur at most twice in w, and by induction w' has the same shorter counts in the window; Theorem A applies. □

Remark 4.3. In case (iii), the matrix form in the proof of Proposition 4.1 gives $|w|_{j+1..l-1} = p+2$, $|w|_{j..l-1} = (p+1)a_1 + a_2$ and $|w|_{j+1..l} = d_1 + (p+1)d_2$, so

$$|w|_{j..l-1}\,|w|_{j+1..l} - |w|_{j..l}\,|w|_{j+1..l-1} = (a_2 - a_1)(d_1 - d_2),$$

the product of the number of letters j between the two letters $j+1$ and the number of letters l between the two letters $l-1$. This 2×2 minor of $\Psi(w)$ is nonnegative, as every minor of a Parikh matrix is [5, Theorem 6, p. 12], [7, Theorem 6, p. 196].

Remark 4.4. Two special cases of Corollary 1.1 are known, and a third result overlaps it. For permutations the second and third diagonals already determine the matrix, and more generally whenever every pair count has Salomaa's γ-property, whatever the multiplicities [9, Theorem 3, p. 226]. Under the hypothesis of Corollary 1.1 the γ-property fails exactly at the edges e with $n_e, n_{e+1} \ge 2$ and $x_e \notin \{0, 1, n_e n_{e+1} - 1, n_e n_{e+1}\}$: at an inner edge this means area 2, where the projection may be $e(e+1)(e+1)e$ or $(e+1)ee(e+1)$, and at an end edge with $n_1 = m \ge 2$ and $n_2 = 2$ (or $n_k = m \ge 2$ and $n_{k-1} = 2$) an area other than $0, 1, 2m-1, 2m$; for instance `43211321` and `14323211`, of content $(3,2,2,1)$, agree on the first three diagonals and have $|w|_{123} = 0$ and 1. For palindromes in which every letter occurs exactly twice, every pair count is 2 and, by (2.1), every longer count is 0 or 2, so the γ-property fails at every edge and crossings do not occur, and the alphabet and the fourth diagonal determine the matrix [13, Theorem 3, Corollary 4, p. 331]; on these words the third diagonal is a function of the second [13, Theorem 3(b)], so Corollary 1.1 contains this result. The general statement and the formulas of Theorem 4.2 were not located in the literature we consulted.

Example 4.5. The words `21332` and `32123` of content $(1,2,2)$ have the same counts of length at most 2 but $|w|_{123} = 0$ and 1; so the fourth diagonal cannot be omitted in Corollary 1.1. For a three-letter window (a,b,c) the corner is determined, for every word of that content, by the counts of length at most 2 if and only if $b = 1$ or $a = c = 1$; this is the case $k = 3$ of [15, Theorem A(iii)]. Sufficiency is also the γ-property [9, Theorem 3]; directly, the corner is $x_1 x_2$ if $b = 1$ (Lemma 2.2(ii)) and $\max(0, x_1 + x_2 - b)$ if $a = c = 1$, and that the corner is not determined in general is noted in [9, p. 226]. For necessity, when $b \ge 2$ and $c \ge 2$ put 2^{b-2} and then 1^{a-1} in front of both words `21332`, `32123` and 3^{c-2} after them; this keeps the shorter counts equal and the corners different. Use the reverse-complement when $a \ge 2$ and $c = 1$. Raising "twice" to "three times" fails as well: Salomaa's construction [9, Lemma 3 and proof of Theorem 4] with $n = 3$ gives the words `1432432111432` and `4321114324321` of content $(4,3,3,3)$, and Theorem B with $n = 3$ gives the words `3221443234` and `2343212443` of content $(1,3,3,3)$; both pairs agree on every count of length at most 3 and have $|w|_{1234} = 1$ and 0. So the five diagonals of Corollary 1.2 cannot be reduced to four.

For the words in which every letter occurs twice, the content (2^k), Corollary 1.1 identifies the Parikh classes, the sets of words with a common Parikh matrix, whose sizes are Salomaa's degrees of ambiguity [12, Definition 6, p. 109].

Corollary 4.6. *On the content* (2^k) *the Parikh classes are exactly the sets of words with given pair counts* $(|w|_{i(i+1)})_i$ *and triple counts* $(|w|_{i(i+1)(i+2)})_i$. *For* $k = 3,\dots,8$ *there are* 30, 173, 993, 5702, 32745 *and* 188045 *such classes, and of the sets of words with given pair counts,* 5, 39, 261, 1577, 9049 *and* 50181 *split into several of them.*

Proof. The first statement is Corollary 1.1; the numbers are computed (Section 8). □

5 A sharpness family

Let $n \geq 2$, let U be the $n \times n$ Ferrers matrix with heights $(1, 2, \ldots, n)$, so $U_{r,s} = [r \leq s]$, and for $1 \leq t \leq n$ let $L_t = U - \mathbf{e}_t \mathbf{e}_t^\top$, the Ferrers matrix with heights $(1, \ldots, t-1, t-1, t+1, \ldots, n)$. As a projection, U is the crossing chain $pqpq \cdots pq$ of two letters, and L_t is the same word with its t-th pair pq inverted.

Lemma 5.1. *For $1 \leq m \leq n$ and $1 \leq a \leq n - m + 1$,*

$$\mathbf{1}^\top L_a L_{a+1} \cdots L_{a+m-1} \mathbf{1} = \binom{n+m}{m+1} - \binom{n+m}{m-1}.$$

Proof. The left side counts the sequences $1 \leq x_0 \leq x_1 \leq \cdots \leq x_m \leq n$ in which no $h \in [m]$ has $x_{h-1} = x_h = a+h-1$. Put $y_h = x_h + h$: these are the $(m+1)$-subsets Y of $[n+m]$, and the forbidden event is $y_{h-1} = a+2h-2$, $y_h = a+2h-1$. Encode Y as a walk $S_0 = 0, S_1, \ldots, S_{n+m}$ with step $+1$ at the elements of Y and -1 elsewhere; it ends at $S_{n+m} = m - n + 2$. Since $\#\{y \in Y : y < a + 2h - 2\} = h - 1$ exactly when $S_{a+2h-3} = 1 - a$, the forbidden event says that the walk crosses the level $\ell = 2 - a$ upwards, from $\ell - 1$ through ℓ to $\ell + 1$, at times $a + 2h - 3, \ldots, a + 2h - 1$; conversely every upward crossing of ℓ has this form, by parity and because $S_t \geq -t$ (parity gives $t - 1 = a + 2h - 3$ for an integer h, $S_{t-1} \geq -(t-1)$ gives $h \geq 1$, and $h \leq m$ because Y then has $h + 1$ elements up to $a + 2h - 1$). So the bad walks are those that visit $\ell - 1$ and later $\ell + 1$, since a walk with steps ± 1 that does so crosses ℓ upwards between the two visits. Since $S_0 = 0 \geq \ell - 1$, reflecting the part before the first visit to $\ell - 1$ maps them bijectively onto the walks from $2\ell - 2$ to $m - n + 2$ that visit $\ell + 1$; as $m - n + 2 \leq \ell + 1$ (that is, $a \leq n - m + 1$), reflecting the part after the first visit to $\ell + 1$ maps these onto all walks from $2\ell - 2$ to $2\ell + 2 - (m - n + 2)$. Their displacement $n - m + 2$ needs $n + 1$ up-steps among $n + m$, so there are $\binom{n+m}{n+1} = \binom{n+m}{m-1}$ bad walks, out of $\binom{n+m}{m+1}$. □

The argument is the reflection principle applied twice; for $a = 1$ the lemma is the classical ballot count of the lattice paths that stay weakly below a diagonal [3, Theorem 10.3.1].

For $m = n - 1$ the value is $\binom{2n-1}{n} - \binom{2n-1}{n+1} = C_n$. In general it is the ballot number $\binom{n+m}{m+1} - \binom{n+m}{m-1} = \frac{n-m}{n+1}\binom{n+m+1}{m+1}$, an entry of Catalan's triangle [6, A009766]; it does not depend on a. We have not found Lemma 5.1 in matrix form, or the words A_n, B_n, in the literature we consulted.

Proof of Theorem B. Over the letters $1, \ldots, n+1$ with content $(1, n, \ldots, n)$, take the projection $(0, \ldots, 0, 1)$ on edge 1 (the letter 1 stands just before the last letter 2), and on edge $i \in \{2, \ldots, n\}$ the heights of L_{i-1} for A_n and of L_i for B_n; the words exist by the realisation remark of Section 2. For $2 \leq i < j \leq n + 1$, (2.1) gives $|A_n|_{i..j} = \mathbf{1}^\top L_{i-1} \cdots L_{j-2} \mathbf{1}$ and $|B_n|_{i..j} = \mathbf{1}^\top L_i \cdots L_{j-1} \mathbf{1}$, both equal to the ballot number of Lemma 5.1 with $m = j - i$. For the counts starting at the letter 1, $\mathbf{1}^\top \mathbf{B}_1 = \mathbf{e}_n^\top$, and $\mathbf{e}_n^\top L_t = \mathbf{e}_n^\top$ for $t < n$ while $\mathbf{e}_n^\top L_n = 0$. Hence $|A_n|_{1..j} = 1$ for $2 \leq j \leq n + 1$, and $|B_n|_{1..j} = 1$ for $j \leq n$ but 0 for $j = n + 1$. □

For $n = 2, 3, 4$ the words are $A_2 = \texttt{32123}$, $B_2 = \texttt{21332}$; $A_3 = \texttt{3221443234}$, $B_3 = \texttt{2343212443}$; and $A_4 = \texttt{32245432132554345}$, $B_4 = \texttt{23432215454323554}$. In Salomaa's construction [9, Theorem 4] the window has the same length and the inner letters the same multiplicity n, but the first letter occurs 2^{n-1} times. In A_n and B_n the last letter occurs n times; we do not know the least possible end multiplicities in general (Section 7).

6 Pivot letters, and the proof of Theorem D

Throughout this section $l - j \geq 4$, and the *pivot* m satisfies $j + 2 \leq m \leq l - 2$, so that $m - 1$, m and $m + 1$ are inner letters of $[j, l]$. For $i \leq m \leq l'$ put

$$f_i = \mathbf{1}^\top \mathbf{B}_i \cdots \mathbf{B}_{m-1}, \qquad g_{l'} = \mathbf{B}_m \cdots \mathbf{B}_{l'-1}\mathbf{1},$$

a row and a column vector of length n_m, with $f_m = \mathbf{1}^\top$ and $g_m = \mathbf{1}$. The r-th entry of f_i is the number of occurrences of $i \cdots (m-1)$ before the r-th letter m, and that of $g_{l'}$ the number of occurrences of $(m+1) \cdots l'$ after it. Let $a = f_{m-1}$, $b = g_{m+1}$, $F = f_j$ and $G = g_l$. So $a_r = A_r^{(m-1)}$ is the number of letters $m-1$ before the r-th letter m, and b_r the number of letters $m+1$ after it. We call (a, b) the *chain* of w at m. (This a is not the a_s of Proposition 4.1.) If $n_m = 0$, every count through m vanishes, so $|w|_{j..l} = 0$ for w and for every word with the same n_m; from now on $n_m \geq 1$.

Lemma 6.1 (Pivot decomposition)**.** (i) *$|w|_{i..l'} = f_i\, g_{l'}$ for $i \leq m \leq l'$. Hence the block $(|w|_{i..l'})_{i \leq m \leq l'}$ of $\Psi(w)$ has rank at most n_m, and*

$$M = \begin{pmatrix} |w|_{j..m} & |w|_{j..m+1} & |w|_{j..l} \\ x_{m-1} & |w|_{m-1..m+1} & |w|_{m-1..l} \\ n_m & x_m & |w|_{m..l} \end{pmatrix} = \sum_{r=1}^{n_m} \begin{pmatrix} F_r \\ a_r \\ 1 \end{pmatrix} \begin{pmatrix} 1 & b_r & G_r \end{pmatrix}.$$

Every entry of M other than $|w|_{j..l}$ is a shorter count of $[j, l]$.

(ii) *$0 \leq a_1 \leq \cdots \leq a_{n_m} \leq n_{m-1}$ and $n_{m+1} \geq b_1 \geq \cdots \geq b_{n_m} \geq 0$.*

(iii) *Let $\varphi = \mathbf{1}^\top \mathbf{B}_j \cdots \mathbf{B}_{m-2}$ and $\gamma = \mathbf{B}_{m+1} \cdots \mathbf{B}_{l-1}\mathbf{1}$. Then $F_r = \varphi_1 + \cdots + \varphi_{a_r}$, and G_r is the sum of the last b_r entries of γ. Hence $F_r = 0$ if $a_r = 0$, $F_r = |w|_{j..m-1}$ if $a_r = n_{m-1}$, and $F_p = F_q$ if $a_p = a_q$; and $G_r = 0$ if $b_r = 0$, $G_r = |w|_{m+1..l}$ if $b_r = n_{m+1}$, and $G_p = G_q$ if $b_p = b_q$.*

Proof. (i) is (2.1) split at the letter m; M is the case $i \in \{j, m-1, m\}$, $l' \in \{m, m+1, l\}$, and $|w|_{j..m+1}$ and $|w|_{m-1..l}$ have lengths $m - j + 2 \leq l - j$ and $l - m + 2 \leq l - j$. (ii) The a_r are the heights of $\mathbf{B}_{m-1}$, and b_r is the number of s with $A_s^{(m)} \geq r$. (iii) $F = \varphi \mathbf{B}_{m-1}$, and the r-th column of $\mathbf{B}_{m-1}$ is the indicator of its first a_r rows. $G = \mathbf{B}_m \gamma$, and the r-th row of $\mathbf{B}_m$ has its ones in the columns s with $A_s^{(m)} \geq r$, which are the last b_r columns since $A^{(m)}$ is nondecreasing. Finally $\varphi\mathbf{1} = |w|_{j..m-1}$ and $\mathbf{1}^\top \gamma = |w|_{m+1..l}$ by (2.1). □

Part (i) says that every occurrence of $i \cdots l'$ passes through exactly one letter m. This is the decomposition with which Mateescu, Salomaa and Yu prove the Cauchy inequality [5, p. 15], taken here for all i and l' at once; for $n_m = 1$ it is the cut rule (Lemma 2.2(ii)). The rank bound is an immediate consequence, but we have not found it stated.

Lemma 6.2 (Letter rectangle rule)**.** *Let $D = x_{m-1}x_m - n_m|w|_{m-1..m+1}$. Then*

$$D = \sum_{1 \leq p < q \leq n_m} (a_q - a_p)(b_p - b_q) \geq 0,$$

and $D = 0$ if and only if a or b is constant. If $D = 0$, then $n_m|w|_{i..l'} = |w|_{i..m}|w|_{m..l'}$ for all $i \leq m \leq l'$.

Proof. By Lemma 6.1(i), $x_{m-1} = \sum_r a_r$, $x_m = \sum_r b_r$ and $|w|_{m-1..m+1} = \sum_r a_r b_r$. Expanding D gives the sum on the right, whose terms are nonnegative by Lemma 6.1(ii). Let $D = 0$ and suppose $a_p \neq a_q$, so that $b_p = b_q$. For every r, if $a_r \neq a_p$ then $b_r = b_p$, and otherwise $a_r \neq a_q$ and $b_r = b_q$; so b is constant. If a is constant, all columns of $\mathbf{B}_{m-1}$ are equal, so $f_i = c_i \mathbf{1}^\top$ for every $i \leq m-1$ and some numbers c_i, and then $|w|_{i..l'} = c_i|w|_{m..l'}$ and $|w|_{i..m} = c_i n_m$. If b is constant, all rows of $\mathbf{B}_m$ are equal, and symmetrically every $g_{l'}$ is a multiple of $\mathbf{1}$. □

Remark 6.3. Lemma 6.2 is known in substance, and we claim nothing new for it. Its first part is [5, Lemma 2, p. 12] for the letters $m-1$, m, $m+1$: equality holds exactly when w contains neither $m(m-1)(m+1)m$ nor $m(m+1)(m-1)m$, which is the condition that a or b be constant. The formula for D is the identity in the proof of the Cauchy inequality [5, p. 15], whose terms are nonnegative when the middle factor is a letter [5, p. 16]. Its second part is the rectangle rule (Lemma 3.1) at edge $m-1$ or m, whose rectangle then contains every letter m.

Lemma 6.4 (Pivot minor)**.** *Let $\Delta = \det M$. By the Cauchy–Binet formula,*

$$\Delta = \sum_{p<q<s} \det \begin{pmatrix} F_p & a_p & 1 \\ F_q & a_q & 1 \\ F_s & a_s & 1 \end{pmatrix} \det \begin{pmatrix} 1 & b_p & G_p \\ 1 & b_q & G_q \\ 1 & b_s & G_s \end{pmatrix}.$$

Hence $\Delta = 0$ if $n_m \le 2$, and also if $n_m = 3$ and two letters m have equal a-values or equal b-values. If $\Delta = 0$ and $D \ne 0$, then

$$D\,|w|_{j..l} = |w|_{j..m}\,x_m\,|w|_{m-1..l} + |w|_{j..m+1}\,x_{m-1}\,|w|_{m..l} - |w|_{j..m}\,|w|_{m-1..m+1}\,|w|_{m..l} - n_m\,|w|_{j..m+1}\,|w|_{m-1..l}. \quad (6.1)$$

Proof. The formula for Δ is the Cauchy–Binet formula for the factorisation of Lemma 6.1(i). For $n_m \le 2$ the sum is empty. For $n_m = 3$ it has one term, and by Lemma 6.1(iii) equal a-values give two equal rows of the first factor, and equal b-values two equal rows of the second. In the expansion of Δ along its first row the coefficient of $|w|_{j..l}$ is D, and (6.1) is the equation $\Delta = 0$. □

The second vanishing statement needs $n_m = 3$: for $n_m \ge 4$ other triples contribute. For instance, the word `43342312515243`, of content $(2,3,4,3,2)$, has the chain $a = (0,0,1,3)$, $b = (2,2,1,0)$ at $m = 3$, and $\Delta = 12$.

Remark 6.5. Δ is the minor of $\Psi(w)$ on the rows $j, m-1, m$ and the columns $m+1, m+2, l+1$, so $\Delta \ge 0$ [5, Theorem 6, p. 12], [7, Theorem 6, p. 196]. There is a classical reason. $\Psi(w)$ is the product of the elementary Jacobi matrices $I + E_{c,c+1}$ (written $x_c(1)$ in [1, p. 6]) over the letters c of w, so it is the weight matrix of the planar network obtained by concatenating their elementary chips [1, p. 7]: levels $1, \dots, k+1$, time running from left to right, and a rising edge from level c to level $c+1$ for each letter c of w. A path from level i to level $l'+1$ is an occurrence of $i \cdots l'$. By Lindström's lemma (the Lindström–Gessel–Viennot lemma) in the form of [1, Lemma 1, p. 2] (see [3, Section 10.13, Theorem 10.13.1 and Corollary 10.13.2] for the general statement and its history, and [2]), every minor of $\Psi(w)$ counts the families of vertex-disjoint paths between the corresponding levels; its nonnegativity is [1, Corollary 2, p. 3]. In this picture, three disjoint paths from the levels j, $m-1$, m to the levels $m+1$, $m+2$, $l+1$ rise from level m to level $m+1$ through three different letters m. Since the paths stay strictly ordered, a letter $m-1$ and a letter $m+1$ lie between any two of these letters m, so their chain is rigid in the sense of Lemma 6.6. This explains every vanishing in Lemma 6.4. We have not found this network used for Parikh matrices, or (6.1) stated; the vanishing statements of Lemma 6.4 are immediate from Lemma 6.1, and we claim nothing for them.

Lemma 6.6 (Chain lemma)**.** *A* chain with bounds (α, β) *is a pair of triples $a = (a_1 \le a_2 \le a_3)$ with values in $\{0, \dots, \alpha\}$ and $b = (b_1 \ge b_2 \ge b_3)$ with values in $\{0, \dots, \beta\}$. Its* signature *is $(\Sigma a, \Sigma b, \Sigma ab)$, where $\Sigma a = a_1 + a_2 + a_3$, $\Sigma b = b_1 + b_2 + b_3$ and $\Sigma ab = a_1b_1 + a_2b_2 + a_3b_3$, and it is* rigid *if $a_1 < a_2 < a_3$ and $b_1 > b_2 > b_3$. If $\alpha, \beta \le 3$, a rigid chain is the only chain with bounds (α, β) that has its signature.*

Proof. A chain with bounds (α, β) also has bounds $(3, 3)$, so we may take $\alpha = \beta = 3$. A rigid a omits exactly one value of $\{0, \dots, 3\}$, so $\Sigma a \in \{3, \dots, 6\}$ determines it, and likewise Σb determines a rigid b. For each $s \in \{3, \dots, 6\}$ there are exactly three nondecreasing triples in $\{0, \dots, 3\}$ with sum s. So a rigid chain has the same $(\Sigma a, \Sigma b)$ as exactly nine chains, itself included, and Table 1 lists Σab for all of them: in each of the 16 blocks the value of the rigid chain occurs once. □

$a\backslash b$	300	210	111	310	220	211	320	311	221	330	321	222
003	0	0	3	0	0	3	0	3	3	0	3	6
012	0	**1**	3	**1**	2	3	**2**	3	4	3	**4**	6
111	3	3	3	4	4	4	5	5	5	6	6	6
013	0	**1**	4	**1**	2	4	**2**	4	5	3	**5**	8
022	0	2	4	2	4	4	4	4	6	6	6	8
112	3	3	4	4	4	5	5	6	6	6	7	8
023	0	**2**	5	**2**	4	5	**4**	5	7	6	**7**	10
113	3	3	5	4	4	6	5	7	7	6	8	10
122	3	4	5	5	6	6	7	7	8	9	9	10
033	0	3	6	3	6	6	6	6	9	9	9	12
123	3	**4**	6	**5**	6	7	**7**	8	9	9	**10**	12
222	6	6	6	8	8	8	10	10	10	12	12	12

Table 1: The values Σab for the nondecreasing triples a (rows) and the nonincreasing triples b (columns) with values in $\{0, \dots, 3\}$ and sums $3, \dots, 6$, in blocks by $(\Sigma a, \Sigma b)$. The rigid chains are bold; in each block their value occurs once.

The lemma fails at the bounds $(3, 4)$, $(4, 3)$ and $(4, 4)$; it still holds at $(2, 4)$ and $(4, 2)$. At bounds $(3, 4)$ the rigid chain $((0, 1, 2), (4, 3, 0))$ has the signature $(3, 7, 3)$, and so do $((0, 0, 3), (3, 3, 1))$ and $((0, 0, 3), (4, 2, 1))$; at bounds $(4, 4)$ there are 100 rigid chains with 95 distinct signatures, 19 of which are shared with another chain.

Lemma 6.7 (Rigid recovery)**.** *Let $n_{m-1}, n_{m+1} \le 3$ and $n_m = 3$, and let the chain (a, b) of w at m be rigid. Then F and G are determined by (a, b) and the counts $|w|_{j..m-1}$, $|w|_{j..m}$, $|w|_{j..m+1}$, $|w|_{m-1..l}$, $|w|_{m..l}$ and $|w|_{m+1..l}$:*

(i) *if $a_1 = 0$, then $F_1 = 0$, $F_2 + F_3 = |w|_{j..m}$ and $b_2F_2 + b_3F_3 = |w|_{j..m+1}$;*

(ii) *otherwise $a = (1, 2, 3)$, $n_{m-1} = 3$, $F_3 = |w|_{j..m-1}$, $F_1 + F_2 = |w|_{j..m} - F_3$ and $b_1F_1 + b_2F_2 = |w|_{j..m+1} - b_3F_3$;*

(iii) *if $b_3 = 0$, then $G_3 = 0$, $G_1 + G_2 = |w|_{m..l}$ and $a_1G_1 + a_2G_2 = |w|_{m-1..l}$;*

(iv) *otherwise $b = (3, 2, 1)$, $n_{m+1} = 3$, $G_1 = |w|_{m+1..l}$, $G_2 + G_3 = |w|_{m..l} - G_1$ and $a_2G_2 + a_3G_3 = |w|_{m-1..l} - a_1G_1$.*

The four linear systems have the nonzero determinants $b_3 - b_2$, $b_2 - b_1$, $a_2 - a_1$ and $a_3 - a_2$.

Proof. By Lemma 6.1(i), $|w|_{j..m} = \sum_r F_r$, $|w|_{j..m+1} = \sum_r b_rF_r$, $|w|_{m..l} = \sum_r G_r$ and $|w|_{m-1..l} = \sum_r a_rG_r$. A rigid a has three distinct values in $\{0, \dots, n_{m-1}\} \subseteq \{0, \dots, 3\}$, so $a_1 = 0$, or $a = (1, 2, 3)$ and $a_3 = n_{m-1} = 3$; by Lemma 6.1(iii), $F_1 = 0$ in the first case and $F_3 = |w|_{j..m-1}$ in the second. The case of b and G is symmetric. □

Theorem 6.8. *Let $l - j \ge 4$ and $j + 2 \le m \le l - 2$, suppose that $n_m \ge 1$, and that $n_m \le 2$ or $n_{m-1}, n_m, n_{m+1} \le 3$, and let D be as in Lemma 6.2.*

(i) *If $D = 0$, then $|w|_{j..l} = |w|_{j..m} |w|_{m..l}/n_m$.*

(ii) *If $D \neq 0$, $n_m = 3$, and $(x_{m-1}, x_m, |w|_{m-1..m+1})$ is the signature of a rigid chain (a, b) with bounds (n_{m-1}, n_{m+1}), then $|w|_{j..l} = F_1G_1 + F_2G_2 + F_3G_3$, where F and G are given by Lemma 6.7 for this chain.*

(iii) *Otherwise $|w|_{j..l}$ is the right side of* (6.1) *divided by D.*

Which case applies is decided by n_{m-1}, n_m, n_{m+1}, x_{m-1}, x_m and $|w|_{m-1..m+1}$, and the right sides involve only shorter counts of $[j, l]$.

Proof. (i) is Lemma 6.2. In (ii), $n_m = 3$, so $n_{m-1}, n_{m+1} \leq 3$ by hypothesis. The chain of w at m has bounds (n_{m-1}, n_{m+1}) by Lemma 6.1(ii) and signature $(x_{m-1}, x_m, |w|_{m-1..m+1})$ by Lemma 6.1(i), so it is the given rigid chain by Lemma 6.6. Then Lemma 6.7 gives F and G, and $|w|_{j..l} = \sum_r F_r G_r$ by Lemma 6.1(i). In (iii), first $\Delta = 0$. If $n_m \leq 2$, this is Lemma 6.4. Otherwise $n_m = 3$, and the chain of w at m is not rigid, for its signature would then be that of a rigid chain with bounds (n_{m-1}, n_{m+1}), which is case (ii); so two of its a-values or two of its b-values coincide, and Lemma 6.4 applies. Since $D \neq 0$, (6.1) gives $|w|_{j..l}$. The counts used are the entries of M other than the corner, which are shorter counts by Lemma 6.1(i), and $|w|_{j..m-1}$ and $|w|_{m+1..l}$, of lengths $m - j$ and $l - m$, both at most $l - j - 2$. □

Proof of Theorem D. Let w' have the same shorter counts as w in the window. It has the same multiplicities, so it satisfies the same hypothesis. If $n_m = 0$, both corners are 0. Otherwise w' also has the same x_{m-1}, x_m and $|w|_{m-1..m+1}$, so the same case of Theorem 6.8 applies to w and w', with the same right side. □

Proof of Corollaries 1.2 and 1.3. The first statement of Conjecture C for $n = 3$ is Theorem D with $m = j + 2$, which is admissible since $l - j \geq 4$. The second follows by strong induction on $l - j$, as in the proof of Corollary 1.1: the counts with $l - j \leq 3$ lie in the first five diagonals, and for $l - j \geq 4$ the inner letters of $[j, l]$ lie in $\{2, \ldots, k - 1\}$. For Corollary 1.3 the same induction applies, with the first hypothesis of Theorem D at the pivot $m = j + 2$, which lies in $\{3, \ldots, k - 2\}$. □

The first hypothesis of Theorem D gives Theorem A again for windows of at least five letters. A four-letter window has no admissible pivot, and there the crossing analysis of Section 4 is needed. The hypothesis of Theorem D is local, while its formulas use counts that reach both ends of the window: for instance, the corner of every word of content $(1, 9, 3, 3, 3, 9, 1)$ is determined by its shorter counts, with pivot $m = 4$.

Theorem D, its explicit form Theorem 6.8 and Corollaries 1.2 and 1.3 were not located in the literature we consulted. Salomaa showed that no fixed number of diagonals suffices in general [9, Theorem 4, p. 228]; Corollary 1.2 shows that five suffice when the letters other than the first and the last occur at most three times, which is the case $n = 3$ of Conjecture C. We were not able to access [11] or [10], which study equality in the Cauchy inequality and the matrices associated with subword histories; we would welcome access to them, and any comments on how they bear on this note, at `christopher.yamba@yambacode.com`.

7 Sharpness, and the conjecture for higher multiplicities

Proposition 7.1. *In each of the following pairs the two words have the stated content over the letters $1, \ldots, k$, and Parikh matrices that differ only in the corner entry, where $|w|_{1..k}$ takes the two stated values. In each five-letter window below, $m = 3$ is the only admissible pivot.*

(i) *The neighbour bounds in Theorem D are needed. The words*

`5432143255555543245`, `4321555543243255554`,

of content $(1,3,3,4,8)$*, have the values* 1 *and* 0*, although of the letters* $m-1$*,* m*,* $m+1$ *only* $m+1=4$ *occurs more than three times; the words*

`543212154321243545`, `123214543255454321`,

of content $(3,4,3,4,4)$*, have the values* 9 *and* 10*. Their reverse complements show the same for the letter* $m-1$*.*

(ii) *The bound* 3 *cannot be raised to* 4 *for the three letters together:* A_4 *and* B_4 *of Theorem B, of content* $(1,4,4,4,4)$*, have the values* 1 *and* 0*. Nor can the pivot alone occur five times: the words*

`32543213325554345`, `23343215555432354`,

of content $(1,3,5,3,5)$*, in which* m *occurs five times and its neighbours three times, have the values* 1 *and* 0*.*

(iii) *A letter that occurs twice must be at distance at least* 2 *from both ends. The words*

`543214325555554345`, `432155554324355554`,

of content $(1,2,3,4,8)$*, in which the letter* 2 *occurs twice, have the values* 1 *and* 0*; in a four-letter window,* `4321324443` *and* `3214443234`*, of content* $(1,2,3,4)$*, have the values* 0 *and* 1*.*

Proof. Direct count; the script `check_c3.py` (Section 8) checks every pair and its reverse complement. □

We do not know whether Theorem D extends to a pivot that occurs four times when its neighbours occur at most three times. The first statement of part (ii) already follows from Salomaa's construction [9, Lemma 3, p. 227, and proof of Theorem 4, pp. 228–229] with $n=4$, whose words have content $(8,4,4,4,4)$; A_4 and B_4 need only one letter 1. The other pairs were found by computer search during the review of this note. None of them contradicts Corollaries 1.2 and 1.3: the four-letter pair differs in a count of length 4, and the five-letter contents satisfy neither hypothesis.

By Theorem B, Conjecture C would be sharp in its bound $l-j \geq n+1$, and by Lemma 2.2(ii), Theorem A and Corollary 1.2 it holds for $n \leq 3$. For $n=4$ the method of Section 6 does not suffice: the chain lemma fails at the bounds $(3,4)$ and $(4,4)$, the anchors of Lemma 6.7 need $n_{m\pm1} \leq 3$, and by Proposition 7.1(i) a five-letter window can be independent when only one of its inner letters occurs four times. (The method of Section 4 fails already at $n=3$: an edge with three letters of each kind can have area 6 with heights $(1,2,3)$, of rank 3, or $(0,3,3)$, of rank 1, so the area no longer tells which rule applies.) The conjecture is a statement about finitely many objects for each window content, since the counts depend only on the projections, and every tuple of projections is realised. We call a window content *independent* if two tuples of projections agree on every count of the window except the corner and differ there, and *determined* otherwise. Table 2 lists the window contents that we checked over all tuples of projections, by hashing the tuples according to their counts other than the corner; up to reversal of the content, which preserves the question by the reverse-complement symmetry (reverse the word and replace each letter i by $k+1-i$, which permutes the counts), each listed content was found determined or, where so marked, independent.

The end letters do not matter for the determination in Theorems A and D, but they do matter for independence below the bound: among the windows $(a,3,3,c)$ with $a,c \leq 6$ the corner is independent exactly when $\max(a,c) \geq 3$, and among $(a,4,4,4,c)$ with $a,c \leq 4$ exactly when $a+c \geq 5$. We have no conjecture for the least end multiplicities.

n	window contents	k	result
2	every content with $n_i \le 2$ and $\sum n_i \le 11$; (2^k), $k \le 8$	4–8	determined (Theorem A)
3	(a, x, y, z, c) with $x, y, z \le 3$, $\max(x, y, z) = 3$, $a, c \le 6$ (357 contents)	5	determined (Corollary 1.2)
3	$(a, 3, 3, 3, 3, c)$ with $a, c \le 4$; (a, x, y, z, t, c) with $x, y, z, t \in \{2, 3\}$, $a, c \le 2$	6	determined (Corollary 1.2)
4	$(1, 4, 4, 4, 4, 1)$, $(1, 4, 4, 4, 4, 2)$, $(2, 4, 4, 4, 4, 2)$, $(1, 4, 4, 4, 4, 4)$, $(2, 4, 4, 4, 4, 3)$	6	determined
4	$(a, 4, 4, 4, c)$ with $a, c \le 4$ and $a + c \ge 5$	5	independent
4	$(a, 4, 4, 4, c)$ with $a, c \le 4$ and $a + c \le 4$	5	determined

Table 2: Exhaustive tests of Conjecture C. The rows with $n = 3$ were computed before Corollary 1.2 was proved, and agree with it. A six-letter window with $n = 4$ is in the range of the conjecture; the contents $(1, 4, 4, 4, 4, 4)$ and $(2, 4, 4, 4, 4, 3)$ are sharp tests, as their five-letter subwindows are independent. The independent five-letter windows are consistent with it, being one letter short.

8 Formal and computer verification

Lean. The file `ParikhCompleteness.lean` (supplementary) formalises the statements of Sections 3 and 4 at the level of words, in Lean 4.34.0 with its core library and no other dependency. Words are lists of natural numbers, $|w|_{j..l}$ is defined by the usual recursion for scattered subwords, and no matrix model is used: the proofs are inductions on the word, following the morphism property of the Parikh matrix one letter at a time, together with splitting lemmas at the first copy of a letter. The formalised theorems are the rectangle rule (Lemma 3.1), the crossing chain, in a parametrised form from which Theorem 4.2(iii) follows by solving two 2×2 systems, the consequence of Lemma 2.1(ii)–(iii) used in the proof (with at most two copies of e and $e + 1$, an edge that is not a crossing avoids $e(e + 1)e(e + 1)$), Theorem A and Corollary 1.1, and, by evaluation in the kernel, Example 4.5 and the corner statement of Theorem B for $n = 3, 4$.

The file `ParikhCompletenessThree.lean` formalises, in the same setting, the determination statements of Section 6: Theorem D (`pivot_theorem`, including the case $n_m = 0$), Corollary 1.2 (`complete_three` and `determined_by_short_three`) and Corollary 1.3 (`determined_by_short_doubled`). The pivot decomposition is proved by inserting a letter m into a word; the identities behind Lemmas 6.2 and 6.4 are polynomial identities over the integers; the chain lemma is checked by evaluation in the kernel over all 4^6 sextuples with entries in $\{0, \dots, 3\}$; and Lemma 6.7 enters as a sixteen-case arithmetic lemma saying that two words with the same rigid chain and the same short counts have the same corner. The explicit formulas of Theorem 6.8 are checked by computer (below), not in Lean.

All these theorems depend only on the standard axioms (propositional extensionality, choice and quotient soundness); the rectangle rule and the chain lemma use propositional extensionality and quotient soundness only, and the evaluated cases of the first file none. There is no `sorry` and no `native_decide`. Lemma 5.1 and Theorem B for general n are proved on paper only.

Computation. The script `check_article3.py` (supplementary, standard-library Python) re-checks: Lemma 3.1 and Theorem 4.2 on every word of every content with $3 \le k \le 5$, $n_i \le 3$ and at most 9 letters; Example 4.5; Lemma 5.1 for $n \le 30$ and all m, a, and Theorem B for $n \le 30$ by building the words and counting; the class numbers of Corollary 4.6; the first row and the five-letter rows of Table 2, and the windows $(a, 3, 3, c)$ of Section 7. The larger entries of Table 2 were computed by the supplementary C program `corner.c`, and most of them were reproduced by an independent Python program written for the review of this note;

neither is part of any proof.

The script `check_c3.py` (supplementary, standard-library Python) re-checks Sections 6 and 7. On every tuple of projections of a list of window contents, and at every admissible pivot, it checks the entries of M, Lemma 6.2, the Cauchy–Binet formula and the sign of Δ, the recovery of Lemma 6.7 against the true F and G, and Theorem 6.8 (465,040 tuples by default and 4,308,830 with `-full`). It also checks every literal word of a few small contents, Lemma 6.6 for all bounds in $\{0,\dots,3\}^2$, Table 1 and the examples after it, the example after Lemma 6.4, Corollaries 1.2 and 1.3 on contents with six and seven letters, the words of Example 4.5 and Proposition 7.1, and Theorem 6.8 on the 27 window contents $(1,x,y,z,1)$ with $x,y,z \le 3$. The following reduction shows that the last check suffices for the second hypothesis. For fixed $\mathbf{B}_{m-1}$ and $\mathbf{B}_m$, the corner $\varphi\,\mathbf{B}_{m-1}\mathbf{B}_m\,\gamma$ and each right side of Theorem 6.8 are bilinear in (φ,γ), since the case depends on $\mathbf{B}_{m-1}$ and $\mathbf{B}_m$ only. The entry φ_s is the number of occurrences of $j\cdots(m-1)$ ending at the s-th letter $m-1$, so φ is nondecreasing, and it is a nonnegative integer combination of the vectors $(0,\dots,0,1,\dots,1)$; dually γ is nonincreasing. In a window with a single letter j and a single letter l, φ and γ run over these vectors, jointly with every $\mathbf{B}_{m-1}$ and $\mathbf{B}_m$, by the realisation remark of Section 2. So Theorem 6.8 holds under its second hypothesis for every window as soon as it holds on those 27 contents. This gives a computer-assisted proof of that case, independent of the proofs of Lemmas 6.4 and 6.7.

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